# CMSC 4200 Artificial Intelligence
# Spring 2026, Group Assignment 5

Due – Tuesday, April 28, 2026, at 11:59 p.m. EST

| Group # | 1 |
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Use this document to provide your answers. This assignment will be graded out of 20 points. Submit your answer file to the right D2L Dropbox named "Assignment 5 Dropbox" by the due date and time.

**Question 1. (6 points)** Use one or more of these predicates *Child*(*x*, *y*), *Sibling*(*x*, *y*), and *Male*(*x*) to define in first-order logic new predicates for:

a) Brother

$\forall_x \forall_y \left(Brother\ (x,y)\right) \Leftrightarrow (Sibling(x,y) \wedge Male(x))$

b) Son

$\forall_x \forall_y \left(Son\ (x,y)\right) \Leftrightarrow (Child(x,y) \wedge Male(x))$

c) First Cousin

*Where (x, y) are first cousins and (p, q) are parents:*

$\forall_{x,y} \left(FirstCousin\ (x,y)\right) \Leftrightarrow \exists_{p,q}(Child(x,p) \wedge Child(y,q) \wedge\ Sibling(p,q))$

**Question 2. (6 points)** Assuming predicates *Parent*(*x*, *y*) and *Female*(*x*) and constants *Alice* and *Peter*, with the obvious meaning, write each of the following sentences in first-order logic:

a) Alice has a daughter (possibly more than one, and possibly sons as well).

$\exists_x\ ((Parent(x, Alice)) \wedge (Female(x))$

b) Alice has exactly one daughter

$\exists_x\ ((Parent(x, Alice) \wedge (Female(x\ )) \ \wedge (\forall_y (Parent(y, Alice)(Female(y\ ) \ \Rightarrow x = y)$

c) Alice and Peter have exactly one child together

$\exists_x\ ((Parent(x, Alice) \wedge \ \left(Parent(x, Peter)\right) \wedge (\forall_y (Parent(y, Alice) \wedge (Parent(y, Peter)) \Rightarrow x = y)$

**Question 3. (3 points)** For each pair of atomic sentences, give the most general unifier if it exists:

a) $P(N, M, z)$, $P(x, y, N)$.
{x -> N, y -> M, z -> N}

b) $Q(x, y, M)$, $Q(N, M, x)$.
Does not exist, since x must be the same in both sentences, but x is forced to be an N in one sentence, and an M in the other sentence. Since x cannot be two things at the same time, we cannot make the sentences equivalent.

c) *Knows(y, y), Knows(Father(x), x)*.
Does not exist, since Knows(y, y) requires both y's to be the same. To make Knows(y, y) = Knows(Father(x), x), that means Father(x) must equal x. This does not have a valid solution because Father(x) cannot equal x, so we cannot make the sentences equivalent.

**Question 4. (5 points)** A FOL knowledge base KB has 5 sentences:

KB = $\{A(Q), \neg A(x) \vee B(x), \neg B(y) \vee C(y), \neg A(z) \vee \neg C(z) \vee D(z), \neg B(w) \vee \neg D(w) \vee E(w)\}$

Does KB entail $E(Q)$? Show your steps.

1.A(Q)

2. $\neg A(x) \vee B(x)$

3. $\neg B(y) \vee C(y)$

4. $\neg A(z) \vee \neg C(z) \vee D(z)$

5. $\neg B(w) \vee \neg D(w) \vee E(w)$

Substitute x for Q using unification

6. $\neg A(Q) \vee B(Q)$

*Use Resolution for 6. Using 1.*

7. $B(Q)$

Substitute y for Q using unification

8. $\neg B(Q) \vee C(Q)$

*Use Resolution for 8. Using 7.*

*9. C(Q)*

Substitute z for Q using unification

10. $\neg A(Q) \vee \neg C(Q) \vee D(Q)$

*Use Resolution for 10. Using 1. And 9.*

*11. D(Q)*

Substitute w for Q using unification

12. $\neg B(Q) \vee \neg D(Q) \vee E(Q)$

*Use Resolution for 12. Using 7. And 11.*

*13. E(Q)*

This shows that KB does entail E(Q)